Non-Instruction Fine-Tuning Data

Artificial intelligence is a field of computer science focused on building systems that can perform tasks associated with human intelligence. Machine learning is a major part of artificial intelligence in which models learn patterns from data rather than relying only on explicitly written rules. A language model learns statistical relationships between tokens by processing large collections of text. During training, the model repeatedly predicts the next token in a sequence and adjusts its parameters to reduce prediction error. A base language model is generally trained on large amounts of raw text and learns knowledge, grammar, vocabulary, syntax, and relationships between concepts. It is different from an instruction-tuned model, which is additionally trained on examples containing instructions and desired responses.

Fine-tuning adapts an existing pretrained model to a particular type of data or domain. In non-instruction fine-tuning, the training data does not need to contain questions, instructions, or answer labels. The model can instead learn directly from continuous domain text. For example, technical documentation, scientific articles, company knowledge, programming documentation, and books can be used as domain text. The training objective can remain causal language modeling, where the model learns to predict the next token from the preceding context.

Parameter-efficient fine-tuning reduces the amount of model parameters that must be updated during training. LoRA is a popular technique that adds small trainable adapter matrices to selected layers while keeping most original model parameters frozen. QLoRA combines LoRA with low-bit quantization so that a large base model can be trained with substantially lower GPU memory usage. Quantization represents model weights using fewer bits, while the adapter parameters remain trainable.

The quality of non-instruction fine-tuning depends strongly on the quality and diversity of the text. Duplicate documents, corrupted text, excessive boilerplate, incorrect information, and very short fragments can reduce training quality. Text is commonly cleaned, normalized, divided into useful chunks, tokenized, and grouped into sequences before training. A validation portion of the data can be used to measure loss on text that was not directly used for optimization.

GPU memory is an important constraint during language-model training. Model weights, activations, gradients, optimizer states, and intermediate tensors all consume memory. Smaller batch sizes, gradient accumulation, mixed-precision computation, gradient checkpointing, and quantization are common techniques for reducing memory requirements. The resulting adapted model can preserve the general capabilities of the base model while becoming more familiar with the vocabulary, terminology, style, and knowledge represented in the domain dataset.